INCH-POUND

MIL-DTL-55302/179C
w/AMENDMENT 1
15 April 2013
SUPERSEDING
MIL-DTL-55302/179C
14 December 2007

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: 4 ROW, PIN ASSEMBLIES, 100 THROUGH 684 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

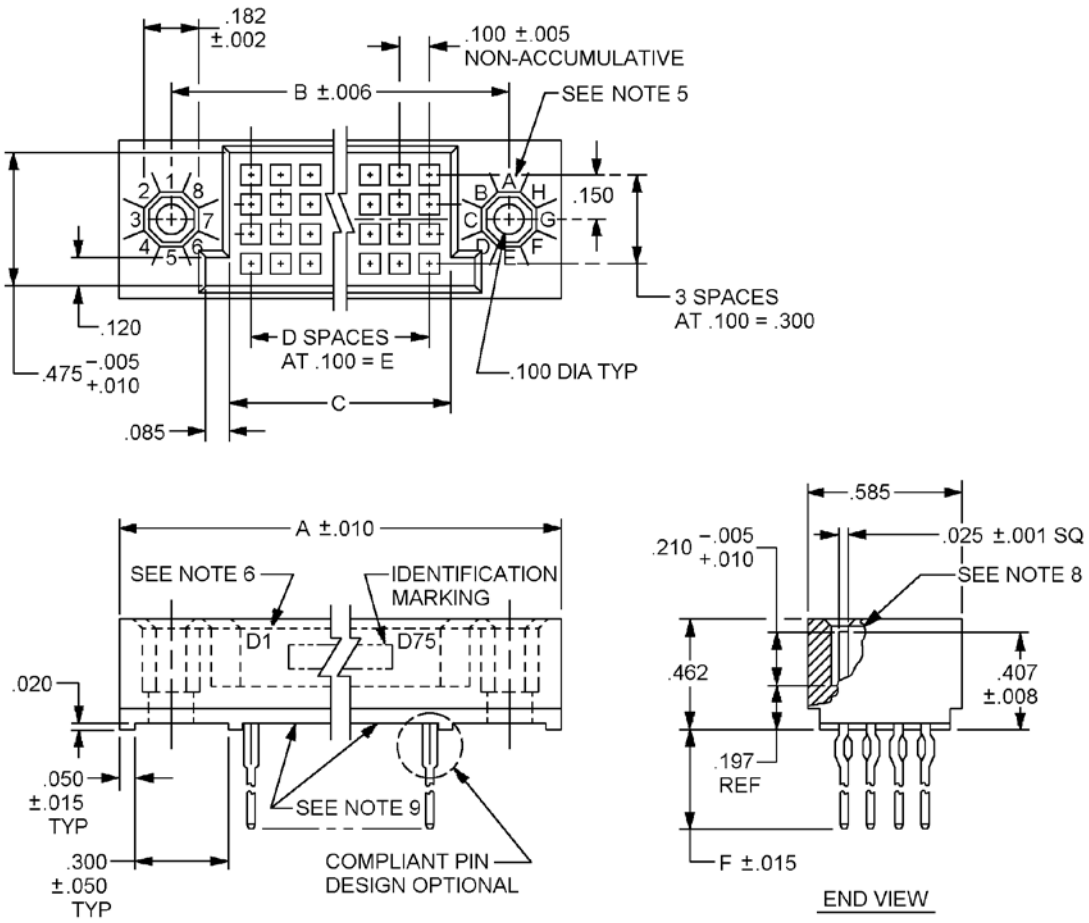
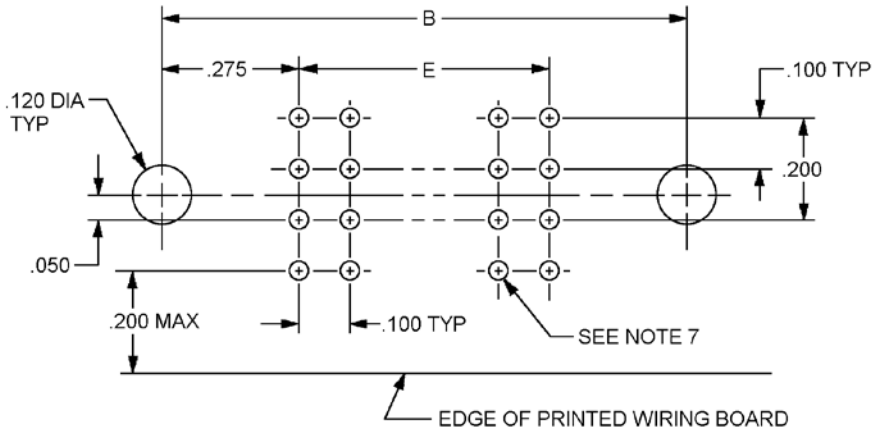


FIGURE 1. Connectors, pin assemblies .100 (2.54 mm) spacing, 100 through 300 contacts.

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Recommended circuit layout

F ± .015 (0.38 mm)		Type	Type of termination
Inches	mm		
.733	18.62	I	Compliant pin (wrappost)
.533	13.54	II	
.250	6.35	IV	
.190	4.83	VI	Solder post (see note 10)

Inches	mm	Inches	mm
.001	0.03	.120	3.05
.002	0.05	.150	3.81
.005	0.13	.182	4.62
.006	0.15	.197	5.00
.008	0.20	.210	5.33
.010	0.25	.250	6.35
.015	0.38	.275	6.98
.020	0.51	.300	7.62
.025	0.64	.407	10.34
.050	1.27	.462	11.73
.085	2.16	.475	12.06
.100	2.54	.585	14.86

FIGURE 1. Connectors, pin assemblies .100 (2.54 mm) spacing, 100 through 300 contacts – Continued.

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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm) on decimals and ± 2° on angles.
4. These connectors mate with connectors specified in MIL-DTL-55302/180.
5. Location indicators embossed on surface.
6. Numbers and indicating end cavities, letters indicating row nearest to polarizing feature, and markings every ten positions stamped on this surface.
7. Holes for solder type shall be .042 - .050 (1.07 – 1.27 mm). Holes for compliant pin shall be manufactured as follows: .0453 ± .0010 (1.151 ± 0.025 mm) drilled hole (1.15 mm drill) .001 - .003 (0.03 – 0.08 mm) thick copper on wall, .0003 (0.008 mm) minimum tin lead plating, finished hole diameter .037 - .043 (0.94 – 1.09 mm) after plating, .036 - .043 (0.91 – 1.09 mm) after reflow, copper hardness (knoop) 150 maximum.
8. Chamfers shall be .015 (0.38 mm) minimum at a 45° angle ± 5°.
9. There shall be a minimum of two standoffs per inch of connector length for the balance of the part. Standoffs shall not compromise the terminating area washout feature.
10. Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0.91 mm) maximum across the diagonal.

FIGURE 1. Connectors, pin assemblies .100 (2.54 mm) spacing, 100 through 300 contacts – Continued.

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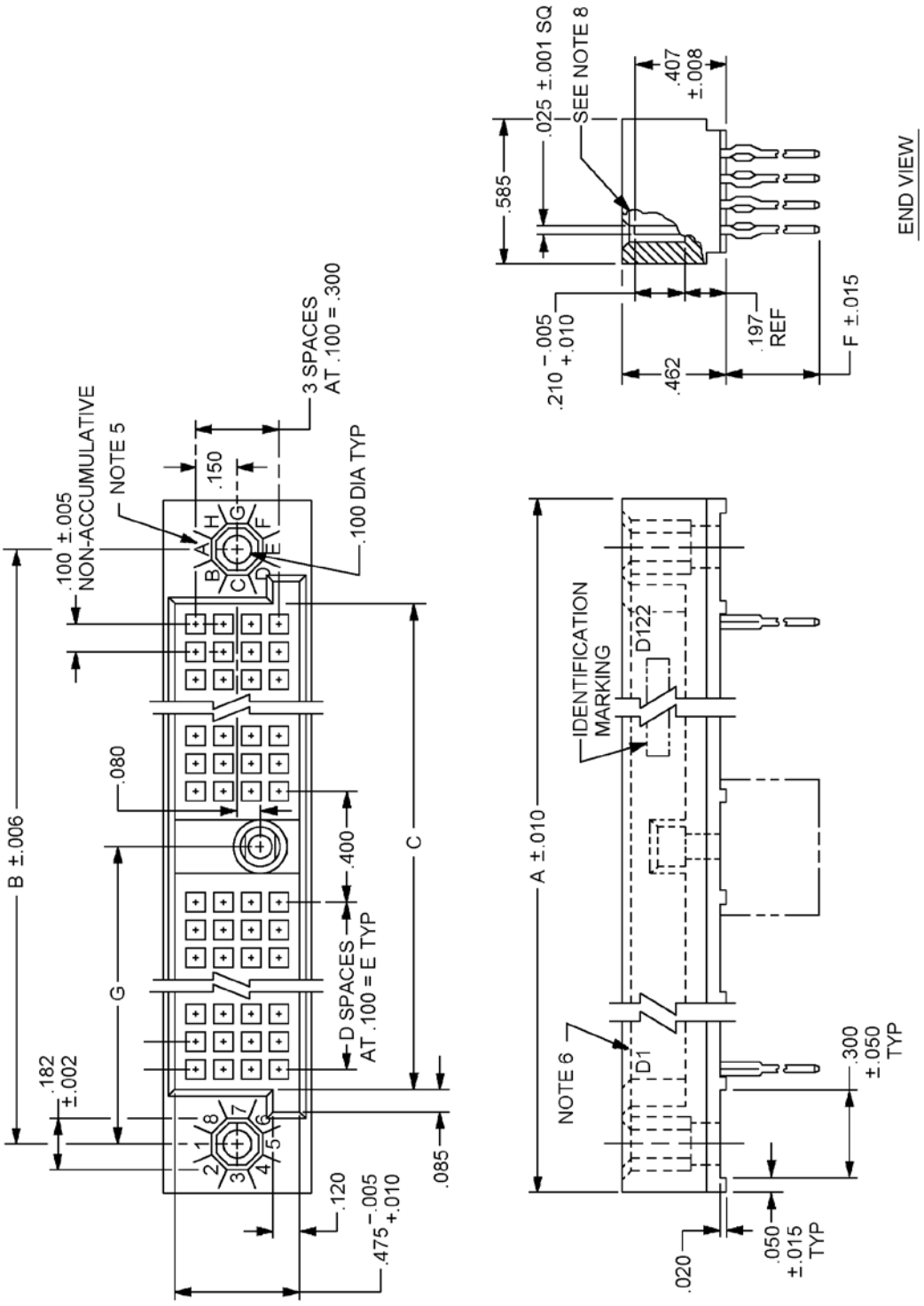
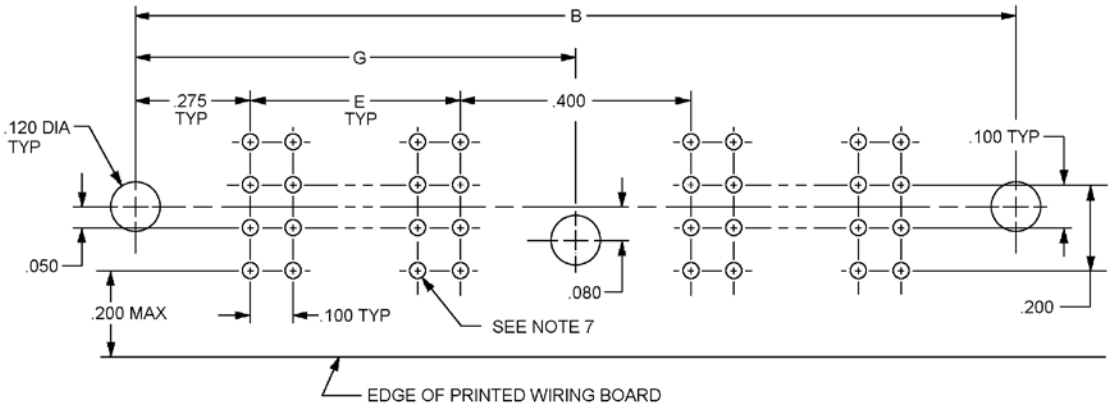


FIGURE 2. Connectors, pin assemblies .100 (2.54 mm) spacing with one guide, 320 through 488 contacts.

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F ± .015 (0.38 mm)		Type	Type of termination
Inches	mm		
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8. Chamfers shall be .015 (0.38 mm) minimum at a 45° angle ± 5°.
9. There shall be a minimum of two standoffs per inch of connector length for the balance of the part. Standoffs shall not compromise the terminating area washout feature.
10. See figure 1 for solder post types. Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0.91 mm) maximum across the diagonal.
11. Guide pins are purchased separately. See MIL-DTL-55302/182.

FIGURE 2. Connectors, pin assemblies .100 (2.54 mm) spacing with one guide, 320 through 488 contacts - Continued.

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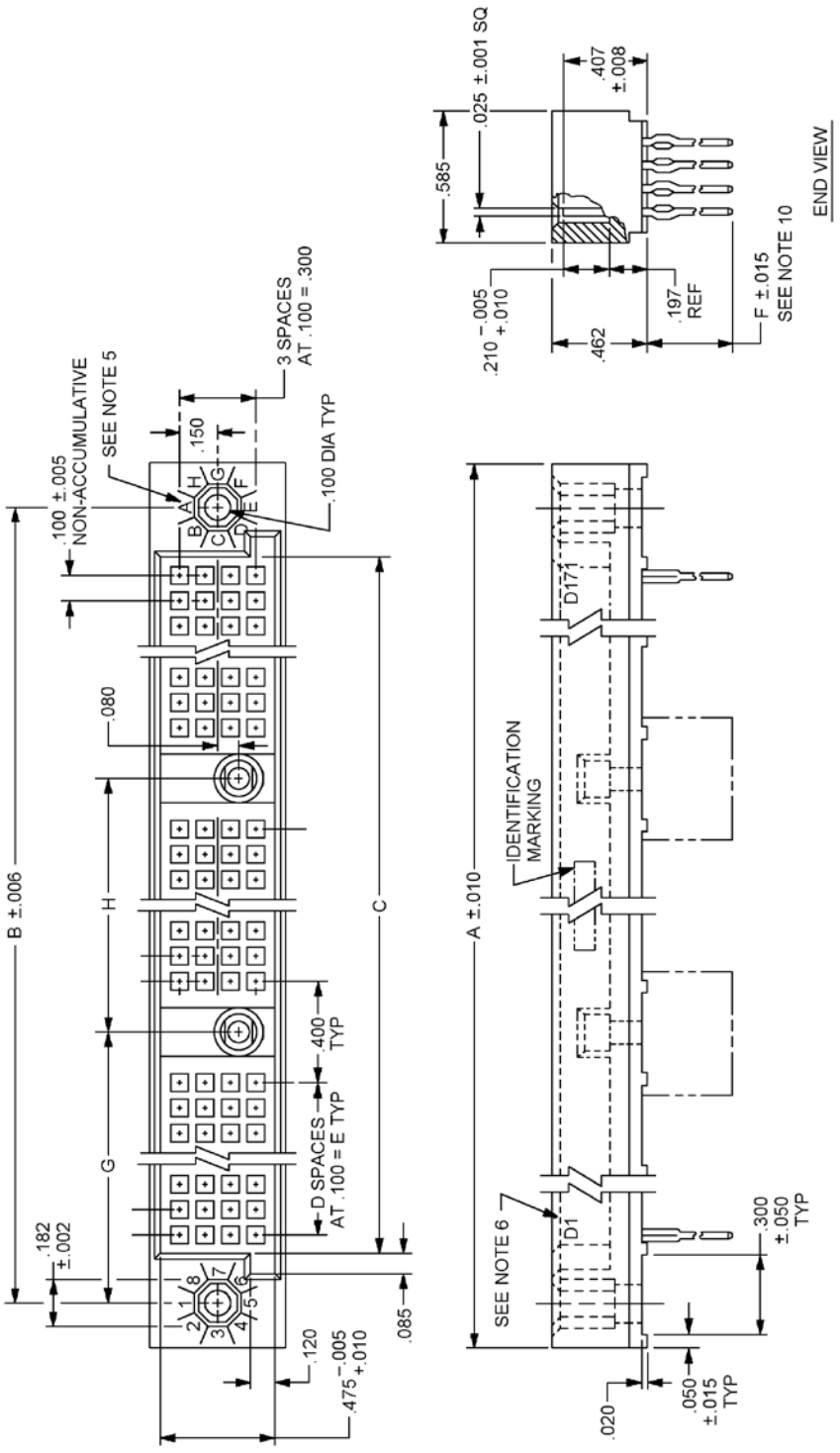
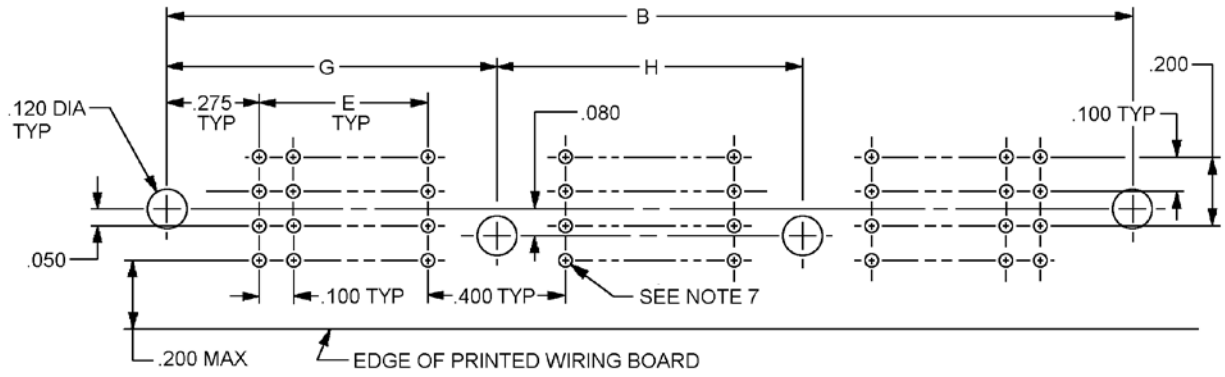


FIGURE 3. Connectors, pin assemblies .100 (2.54 mm) spacing with two guides, 516 through 684 contacts.

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F ± .015 (0.38 mm)		Type	Type of termination
Inches	mm		
.733	18.62	I	Compliant pin (wrappost)
.533	13.54	II	
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.010	0.25	.150	3.81	.475	12.06
.015	0.38	.182	4.62	.585	14.86
.020	0.51	.197	5.00		
.025	0.64	.210	5.33		

FIGURE 3. Connectors, pin assemblies .100 (2.54 mm) spacing with two guides, 516 through 684 contacts - Continued.

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4. These connectors mate with connectors specified in MIL-DTL-55302/180.
5. Location indicators embossed on surface.
6. Numbers and indicating end cavities, letters indicating row nearest to polarizing feature, and markings every ten positions stamped on this surface.
7. Holes for solder type shall be .042 - .050 (1.07 – 1.27 mm). Holes for compliant pin shall be manufactured as follows: .0453 ± .0010 (1.151 ± 0.025 mm) drilled hole (1.15 mm drill) .001 - .003 (0.03 – 0.08 mm) thick copper on wall, .0003 (0.008 mm) minimum tin lead plating, finished hole diameter .037 - .043 (0.94 – 1.09 mm) after plating, .036 - .043 (0.91 – 1.09 mm) after reflow, copper hardness (knoop) 150 maximum.
8. Chamfers shall be .015 (0.38 mm) minimum at a 45° angle ± 5°.
9. There shall be a minimum of two standoffs per inch of connector length for the balance of the part. Standoffs shall not compromise the terminating area washout feature.
10. Solder post shall be a .036 (0.91 mm) maximum diameter or .036 (0.91 mm) maximum across the diagonal.
11. Guide pins are purchased separately. See MIL-DTL-55302/182.

FIGURE 3. Connectors, pin assemblies .100 (2.54 mm) spacing with two guides, 516 through 684 contacts - Continued.

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TABLE I. Dash number, number of contacts, contact type, and dimensions. 1/ 2/

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
01 02 04 06	100	1	I II IV VI	3.250 (82.55)	2.950 (74.93)	2.550 (64.77)	24	2.400 (60.96)			1-25
19 20 22 24	128	1	I II IV VI	3.950 (100.33)	3.650 (92.71)	3.250 (82.55)	31	3.100 (78.74)			1-32
37 38 40 42	160	1	I II IV VI	4.750 (120.65)	4.450 (113.03)	4.050 (102.87)	39	3.900 (99.06)			1-40
55 56 58 60	200	1	I II IV VI	5.750 (146.05)	5.450 (138.43)	5.050 (128.27)	49	4.900 (124.46)			1-50
64 65 67 69	220	1	I II IV VI	6.250 (158.75)	5.950 (151.13)	5.550 (140.97)	54	5.400 (137.16)			1-55
73 74 76 78	240	1	I II IV VI	6.750 (171.45)	6.450 (163.83)	6.050 (153.67)	59	5.900 (149.86)			1-60
82 83 85 87	260		I II IV VI	7.250 (184.15)	6.950 (176.53)	6.550 (166.37)	64	6.400 (162.56)			1-65
91 92 94 96	268		I II IV VI	7.450 (189.23)	7.150 (181.61)	6.750 (171.45)	66	6.600 (167.64)			1-67
100 101 103 105	280	1	I II IV VI	7.750 (196.85)	7.450 (189.23)	7.050 (179.07)	69	6.900 (175.26)			1-70

See footnote at end of table.

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TABLE I. Dash number, number of contacts, contact type, and dimensions - Continued. 1/ 2/

Dash no.	No. of contacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
109 110 112 114	300	1	I II IV VI	8.250 (209.55)	7.950 (201.93)	7.550 (191.77)	74	7.400 (187.96)			1-75
127 128 130 132	344	2	I II IV VI	9.650 (245.11)	9.350 (237.49)	8.950 (227.33)	42	4.200 (106.68)	4.675 (118.75)		1-86
154 155 157 159	416	2	I II IV VI	11.450 (290.83)	11.150 (283.21)	10.750 (273.05)	519	5.100 (129.54)	5.575 (141.61)		1-104
181 182 184 186	488	2	I II IV VI	13.250 (336.55)	12.950 (328.93)	12.550 (318.77)	60	6.000 (152.40)	6.475 (164.47)		1-122
190 191 193 195	516	3	I II IV VI	14.250 (361.95)	13.950 (354.33)	13.550 (344.17)	42	4.200 (106.68)	4.675 (118.75)	4.600 (116.84)	1-129
199 200 202 204	540	3	I II IV VI	14.850 (377.19)	14.550 (369.57)	14.150 (359.41)	44	4.400 (111.76)	4.875 (123.83)	4.800 (121.92)	1-135
226 227 229 231	612	3	I II IV VI	16.650 (422.91)	16.350 (415.29)	15.950 (405.13)	50	5.000 (127.00)	5.475 (139.07)	5.400 (137.16)	1-153
235 236 238 240	636	3	I II IV VI	17.250 (438.15)	16.950 (430.53)	16.550 (420.37)	52	5.200 (132.08)	5.675 (144.15)	5.600 (142.24)	1-159
253 254 256 258	684	3	I II IV VI	18.450 (468.63)	18.150 (461.01)	17.750 (450.85)	56	5.600 (142.24)	6.075 (154.31)	6.000 (152.40)	1-171

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2/ Metric equivalents are given for information only.

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TABLE II. Contact plating requirements. 1/

Type	Finish requirement
A	Overall contact finish: Shall be gold in accordance with MIL-DTL-45204, class 1, type II, grade C or equivalent.
B	Localized contact finish. Contact engagement area, .170-inch minimum length, shall be gold over nickel in accordance with MIL-DTL-45204, class 1, type II, grade C or equivalent. Compliant pin area (contact type I, .748 inch minimum length, type II, .548 minimum length; type IV, .265 inch minimum length) shall be tin-lead over nickel in accordance with MIL-DTL-55302. Solder post area contact type VI, .205 minimum length, shall be tin-lead over nickel in accordance with MIL-DTL-55302.

1/ Type C finish is deleted and superseded by type A or B.

REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figures 1, 2, 3 and table I.

Materials: Shall be in accordance with MIL-DTL-55302.

Plating: See table II.

Current rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient with no more than two adjacent contacts carrying this current.

Keying (see MIL-DTL-55302/31):

Two keys, Part or Identifying Number (PIN) M55302/31-04, and two .086-56 UNC-2A mounting screws are recommended. Keys and screws shall be ordered separately.

Jackscrews (see MIL-DTL-55302/182):

Use M55302/182-11, -12, -13, -14, -15, or –16. Jackscrews are purchased separately.

Mating and unmating: The mating force in pounds shall be the number of contacts multiplied by .250; the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding .015-ohm initial and .020 after testing.

Contact retention: Not applicable.

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Dielectric withstanding voltage:

Sea level: 900 volts rms.

70,000 feet: 200 volts rms.

2 milliamperes maximum leakage current.

PIN: M55302/179 (and dash number from table I and type from table II).

Group qualification: See table III.

TABLE III. Group qualification.

Qualification of any of the following connectors 1/ 2/	Qualifies the following connectors	Retains qualification on the following
M55302/173-*** M55302/175-***	M55302/173-*** M55302/175-***	M55302/173-*** M55302/175-*** M55302/176-*** M55302/178-*** M55302/179-*** M55302/181-***
M55302/176-*** M55302/178-***	M55302/173-*** M55302/175-*** M55302/176-*** M55302/178-***	
M55302/179-*** M55302/181-***	M55302/173-*** M55302/175-*** M55302/176-*** M55302/178-*** M55302/179-*** M55302/181-***	

1/ For initial qualification, connectors with the largest number of contact positions from the left hand column shall be used to obtain qualification for parts in corresponding center column.
2/ For qualification retention, data may be supplied on any two parts from the left hand column in order to retain qualification for the parts listed in the right hand column.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/31
MIL-DTL-55302/180
MIL-DTL-55302/182
MIL-DTL-45204

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy - EC
Air Force – 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2013-067)

Review activities:
Army – AR, AT, AV, CR4, MI
Navy – AS, MC, SH
Air Force - 19, 99

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